

Metal-Organic Cages

Length Ratio-Driven Configurational Modulation of Heteroleptic $\text{Pd}_6\text{L}_6\text{L}'_6$ Cages

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Abstract: Biomolecular configurational modulation is essential for maintaining the stability and functionality of biological systems. Drawing inspiration from this natural phenomenon, we present a strategy to modulate the configurations of heteroleptic $\text{Pd}_6\text{L}_6\text{L}'_6$ cages by systematically varying the ligand length ratio (r). Although theoretical analysis suggests that these $\text{Pd}_6\text{L}_6\text{L}'_6$ assemblies could adopt up to 48 possible configurational isomers, we successfully modulated the thermodynamic preference between two dominant configurations (Type-A and Type-B) by implementing energetic differentiation through careful optimization of r . Specifically, the system exclusively forms Type-A cages at $r = 1.0$, whereas Type-B becomes the predominant configuration when $r \geq 1.4$. Control experiments demonstrate that the introduction of bulky steric groups does not induce configuration changes. This research offers valuable insights and serves as a useful reference for the design and configurational modulation of heteroleptic cages.

Introduction

The precise modulation of biomolecular configurations serves as a foundational mechanism for maintaining structural integrity and functional specificity in living systems.^[1–3] For instance, variations in DNA helix handedness and disulfide bond rearrangement exert influence on cellular processes.^[4–6] These ingenious natural mechanisms have inspired chemists to create artificial systems that mimic such biological behaviors.^[7–10] Among diverse artificial supramolecular architectures, the metal-organic cages (MOCs) constructed by “naked” Pd(II) or Pt(II) ions have become an ideal choice

for emulating biomolecular regulation, due to their reversible and moderately stable metal coordination bonds.^[11–16]

Currently, research on the configurational modulation of MOCs primarily focuses on the homoleptic cages constructed using asymmetric ligands.^[17,18] Due to the inherent asymmetry of these ligands, the assembled products often exhibit multiple isomers.^[19,20] A case in point is Pd_2L_4 , which exhibits four configurations (Figure 1a). Through effective steric or geometric strategies, researchers have achieved precise control over *all-up*, *cis* and *trans* configurations.^[21,22] Similarly, advancements have been made in the regulation of higher-nuclearity cages.^[23–26] Nonetheless, reports on configuration modulation of heteroleptic cages are exceedingly scarce.^[27–29] Three fundamental challenges persist (Figure 1b): (1) circumventing the narcissistic self-sorting tendency of ligands is imperative;^[30–33] (2) suppressing the formation of statistical mixtures;^[34–36] (3) the more central issue is how to precisely tune the inherent thermodynamic properties or stability of different configurations. Thus far, some adjustments to the *cis/trans* configurations of $\text{Pd}_2\text{L}_2\text{L}'_2$ have been documented.^[27] In the context of high-nuclearity heteroleptic cages, as the nuclearity (n) of the cage structure increases, the number of potential isomers increases exponentially.^[37–39] For instance, in the cuboctahedral cage $\text{Pd}_{12}\text{L}_{12}\text{L}'_{12}$ reported by Fujita, theoretically, up to 113048 isomers can be generated if the two ligands are randomly assigned to the edges of the cuboctahedral topology.^[40] This underscores the immense challenge of precisely directing and synthesizing single-configuration products of high-nuclearity cages.^[41]

Herein, we propose a strategy for achieving configurational modulation of heteroleptic cages by adjusting the ligand length ratio r (defined as $r = d_i/d_1$, where $i = 2, 3, 4, 5, 6$; d represents the distance between the two nitrogen coordinating atoms in the ligand). By assembling ditopic ligand **L1** and different ligands **L2–L6**, we have successfully constructed a series of heteroleptic octahedral $\text{Pd}_6\text{L}_6\text{L}'_6$ cages. Single-crystal structure analysis shows that by varying the ligand length ratio r ($r = d_{2,6}/d_1$) of these two types of ligands, the energy differences between distinct configurations can be finely tuned, thereby achieving modulation from Type-A to Type-B among the possible 48 configurational isomers (Figure 1c). Specifically, when r is 1.0 (**L**²/**L**¹ and **L**⁵/**L**¹), the resulting $\text{Pd}_6\text{L}_6\text{L}'_6$ cages adopt the Type-A configuration. As r is sequentially increased to 1.4, 1.5, and 1.7 (corresponding to **L**³/**L**¹, **L**⁴/**L**¹ and **L**⁶/**L**¹, respectively), the cage configuration shifts from Type-A to Type-B. Notably, the introduction of sterically bulky substituents on the ligands (**L**⁵ and **L**⁶)

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Additional supporting information can be found online in the
 Supporting Information section

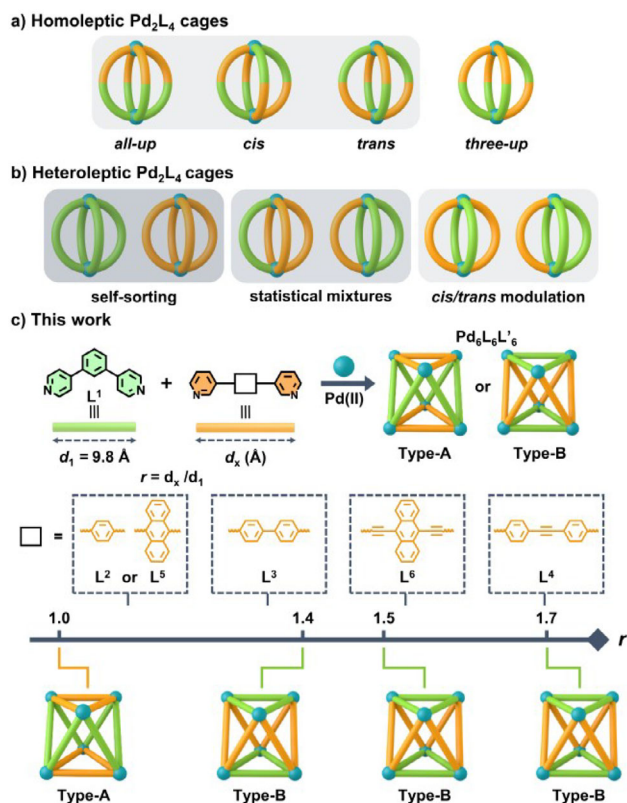


Figure 1. a) Potential isomers of Pd_2L_4 cages assembled from asymmetric ligands. b) The six possible structures that may arise when the two types of ligands are assembled. c) The configuration modulation of $\text{Pd}_6\text{L}_6\text{L}'_6$ cages is achieved in this study.

did not compromise the ligand length ratio-controlled cage configuration modulation. This work provides a versatile and predictable approach for controlling configuration through simple geometric parameters.

Results and Discussion

The ligands $\text{L}^1\text{--L}^6$ were synthesized through efficient Suzuki and Sonogashira coupling reactions, with detailed synthetic routes and characterizations provided in the Supporting Information (Scheme S1, Figures S1–S4 and S25). The heteroleptic $\text{Pd}_6\text{L}_6\text{L}'_6$ cage has been successfully screened and obtained by Severin research group from six distinct dipyrindyl ligands mixed with $[\text{Pd}(\text{CH}_3\text{CN})_4](\text{BF}_4)_2$ (Figure 2a).^[42] Geometric analysis reveals that when two types of ligands are randomly distributed along the 12 edges of the cage, 48 distinct configurations arise (see the Supporting Information for further details). However, it is noteworthy that the heteroleptic $\text{Pd}_6\text{L}_6\text{L}'_6$ cage exhibits extremely high selectivity, exclusively producing the Type-A configuration, indicating that the energy of the Type-A configuration is significantly lower than that of other configurations. As shown in Figure 2b, the angles of the triangle formed by Pd_1 , Pd_2 , and Pd_3 are 53° , 63° , and 64° respectively, which are close to the 60° of an equilateral triangle, effectively alleviating the spatial congestion between

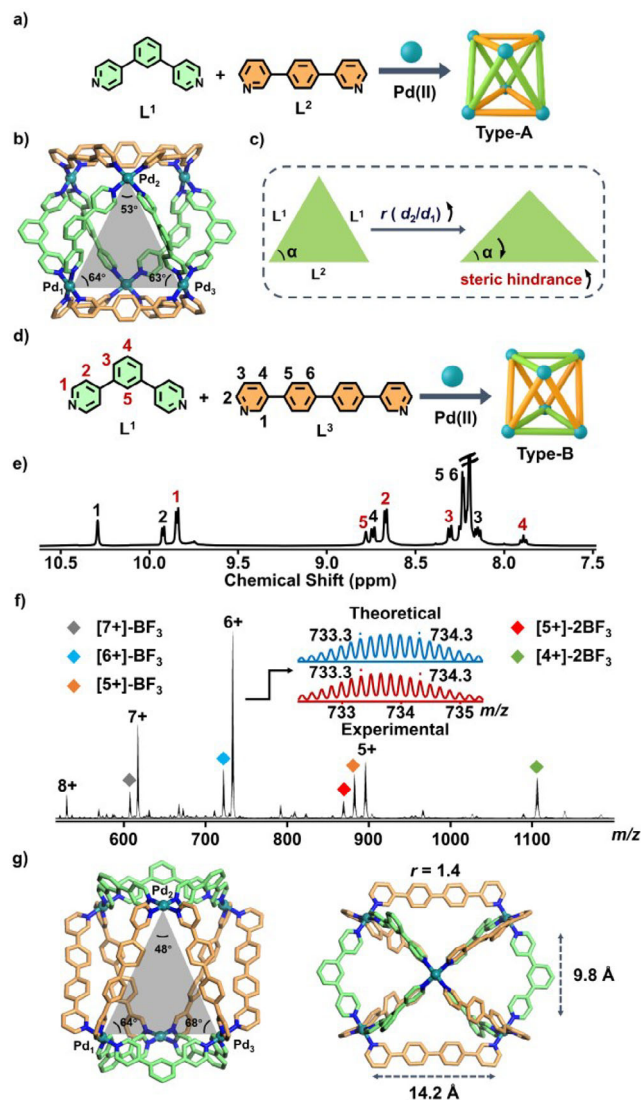


Figure 2. a) The assembled complex of L^1 , L^2 and $\text{Pd}(\text{II})$ adopts the Type-A configuration. b) The single-crystal structure of $\text{Pd}_6\text{L}_6\text{L}'_6$.^[42] c) Adjustment of length ratio r induces steric hindrance variation. d) The assembled complex of L^1 , L^3 , and $\text{Pd}(\text{II})$ adopts the Type-B configuration. e) ^1H NMR spectrum (500 MHz, 300 K) of $\text{Pd}_6\text{L}_6\text{L}'_6$. f) ESI-MS spectrum of $\text{Pd}_6\text{L}_6\text{L}'_6$. g) The single-crystal structure of $\text{Pd}_6\text{L}_6\text{L}'_6$ (hydrogen atoms and disorder have not been shown for clarity).^[44]

ligands. We hypothesize that increasing r of L^2 ($d_2 = 9.9 \text{ \AA}$) to L^1 ($d_1 = 9.8 \text{ \AA}$) will cause the triangle angles to deviate from 60° , i.e., $\alpha < 60^\circ$ (Figure 2c), thus intensifying the spatial congestion among ligands. This alteration may impact the energy differences between various configurations, leading to alterations in the optimal configuration. Given our previous success in controlling the configuration of low-symmetry Pd_2L_4 cages by adjusting the length ratio of ligand arms,^[43] we now decide to explore the feasibility of this strategy in modulating the configurations of heteroleptic $\text{Pd}_6\text{L}_6\text{L}'_6$ cages.

After introducing an additional phenyl spacer to elongate the L^2 ligand, we obtained ligand L^3 ($d_3 = 14.2 \text{ \AA}$) with $r_2 = 1.4$. L^1 , L^3 , and $[\text{Pd}(\text{CH}_3\text{CN})_4](\text{BF}_4)_2$ were dissolved

in DMF-*d*₇ solvent in a precise 1:1:1 stoichiometric ratio and allowed to react at 70 °C overnight to obtain the final assembly product. The ¹H NMR spectrum exhibited a clear set of proton signals (Figure 2e), with the proton signals adjacent to the pyridine N-atoms appearing at lower field positions, indicating successful assembly. Due to the loss of different amounts of anions, a series of peaks with consecutive charge were detected by electrospray ionization mass spectrometry (ESI-MS) (Figure 2f). After deconvolution of these peaks, the exact molecular weight was accurately calculated to be 4922.8 Da, which matches the composition of six **L**¹, six **L**³, six Pd(II) ions, and twelve BF₄[−], thus confirming the molecular composition of the assembled product as [Pd₆(**L**¹)₆(**L**³)₆](BF₄)₁₂, named **Pd₆L¹₆L³₆**. Notably, both ligands showed only one type of proton signal corresponding to a single chemical environment in the ¹H NMR spectrum, indicating that **Pd₆L¹₆L³₆** has high symmetry. Through screening, we found that, among the 48 conformational species, apart from the Type-A configuration, the Type-B configuration also exhibits the highest level of symmetry. To confirm the configuration of **Pd₆L¹₆L³₆**, crystal growth experiments were conducted. By slowly evaporating tetrahydrofuran into a DMF solution of **Pd₆L¹₆L³₆** over three weeks, we successfully obtained plate-like crystals suitable for testing. As shown in Figure 2g, unlike the Type-A cage **Pd₆L¹₆L²₆**, the obtained crystal structure where two macrocyclic [**Pd₃L¹₃**]⁶⁺ fragments are bridged by six **L**³ ligands, corresponds to Type-B. The angles of the triangle formed by Pd₁, Pd₂, and Pd₃ are 48°, 64°, and 68°, respectively. Through comprehensive characterization using NMR, MS, and single-crystal, it has been initially demonstrated that the configuration can be thermodynamically regulated from Type-A to Type-B by adjusting *r*.

To further validate that the configurational change was indeed caused by variations in *r*, we designed the ligand **L**⁴ (*d*₄ = 16.7 Å) with *r*₃ = 1.7. By using the same assembly method as **Pd₆L¹₆L³₆**, we obtained the co-assembly product of **L**¹ and **L**⁴. Through performing integral analysis on the proton signals in the ¹H-NMR spectrum, the stoichiometric ratio of the two ligands, **L**¹ and **L**⁴, in the co-assembled product was confirmed to be exactly 1:1 (Figure 3a, Figure S11). As shown in Figure 3b, ESI-MS analysis validated the molecular composition as [Pd₆(**L**¹)₆(**L**⁴)₆](BF₄)₁₂, named **Pd₆L¹₆L⁴₆**. However, NMR and MS alone are insufficient to determine its configuration. Fortunately, by slowly diffusing tetrahydrofuran vapor into a DMF solution of **Pd₆L¹₆L⁴₆**, we successfully grew crystals suitable for structural testing. Due to the incorporation of the alkyne spacer, the rigidity of **L**⁴ was weakened, resulting in slight bending within the crystal structure. Meanwhile, the configuration of **Pd₆L¹₆L⁴₆** was consistent with that of **Pd₆L¹₆L³₆**, both belonging to Type-B (Figure 3c). The angles of the triangle formed by Pd₁, Pd₂, and Pd₃ are 42°, 68°, and 70°, respectively. Therefore, it is straightforward to infer that the configuration of **Pd₆L¹₆L⁶₆** is influenced by the ligand length ratio.

Based on the above experiments, the ligand length ratio has been identified as a crucial factor in modulating the configuration. When the *r*-value deviates from 1,

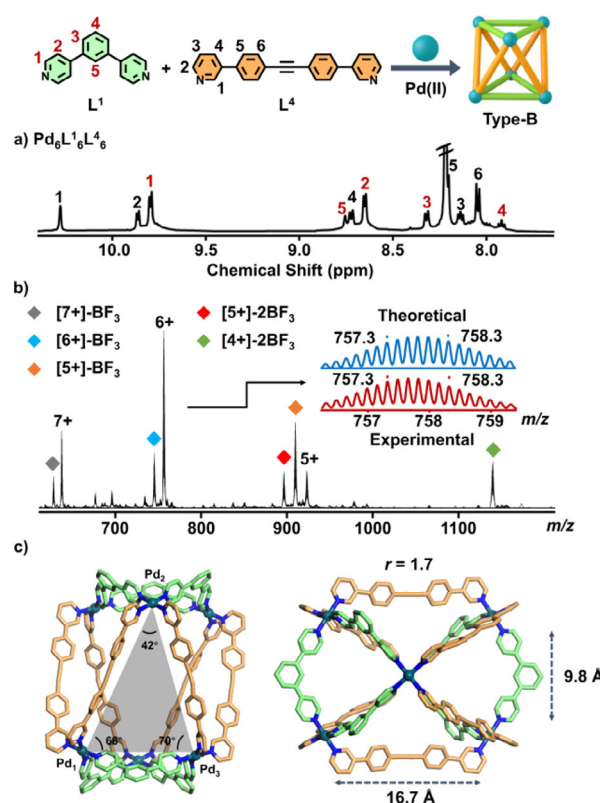


Figure 3. The assembled complex of **L**¹, **L**⁴, and Pd(II) adopts the Type-B configuration. a) ¹H NMR spectrum (500 MHz, 300 K) of **Pd₆L¹₆L⁴₆**. b) ESI-MS spectrum of **Pd₆L¹₆L⁴₆**. c) The single-crystal structure of **Pd₆L¹₆L⁴₆** (hydrogen atoms and disorder have not been shown for clarity).^[44]

the steric hindrance effects between neighboring ligands are significantly enhanced, which consequently adjusts the energy disparity between Type-A and Type-B configurations, thereby enabling configuration modulation. Subsequently, we investigated whether the introduction of sterically bulky groups onto the ligands would affect the configurational modulation induced by changes in *r*. To this end, we introduced anthryl groups to further explore their generality. Ligands **L**⁵ (*d*₅ = 9.9 Å, *r*₄ = 1.0) and **L**⁶ (*d*₆ = 15.1 Å, *r*₅ = 1.5) were designed, which were significantly increased steric bulk. Firstly, a co-assembly experiment with **L**¹ and **L**⁵ was conducted. With a precisely controlled stoichiometric ratio of 1:1:1, **L**¹, **L**⁵, and Pd(CH₃CN)₄(BF₄)₂ were dissolved in DMF-*d*₇ and reacted at 70 °C for 8 h, yielding an ordered assembly product. This result was verified by ¹H NMR (Figure 4a), and ESI-MS also detected signals matching the hexanuclear complex (Figure 4b), confirming the formation of [Pd₆(**L**¹)₆(**L**⁵)₆](BF₄)₁₂, which we named **Pd₆L¹₆L⁵₆**. Since both **L**⁵ and **L**² have a length ratio of 1.0 with **L**¹, it is inferred that **Pd₆L¹₆L⁵₆** should adopt a Type-A configuration. To verify this, a crystal growth experiment was conducted. By slowly diffusing isopropyl ether vapor into a DMSO solution of **Pd₆L¹₆L⁵₆**, yellow crystals were successfully cultivated. Crystallographic analysis confirmed our inference that **Pd₆L¹₆L⁵₆** indeed possesses a Type-A configuration (Figure 4c), similar to that of **Pd₆L¹₆L²₆**, with six **L**¹ acting as

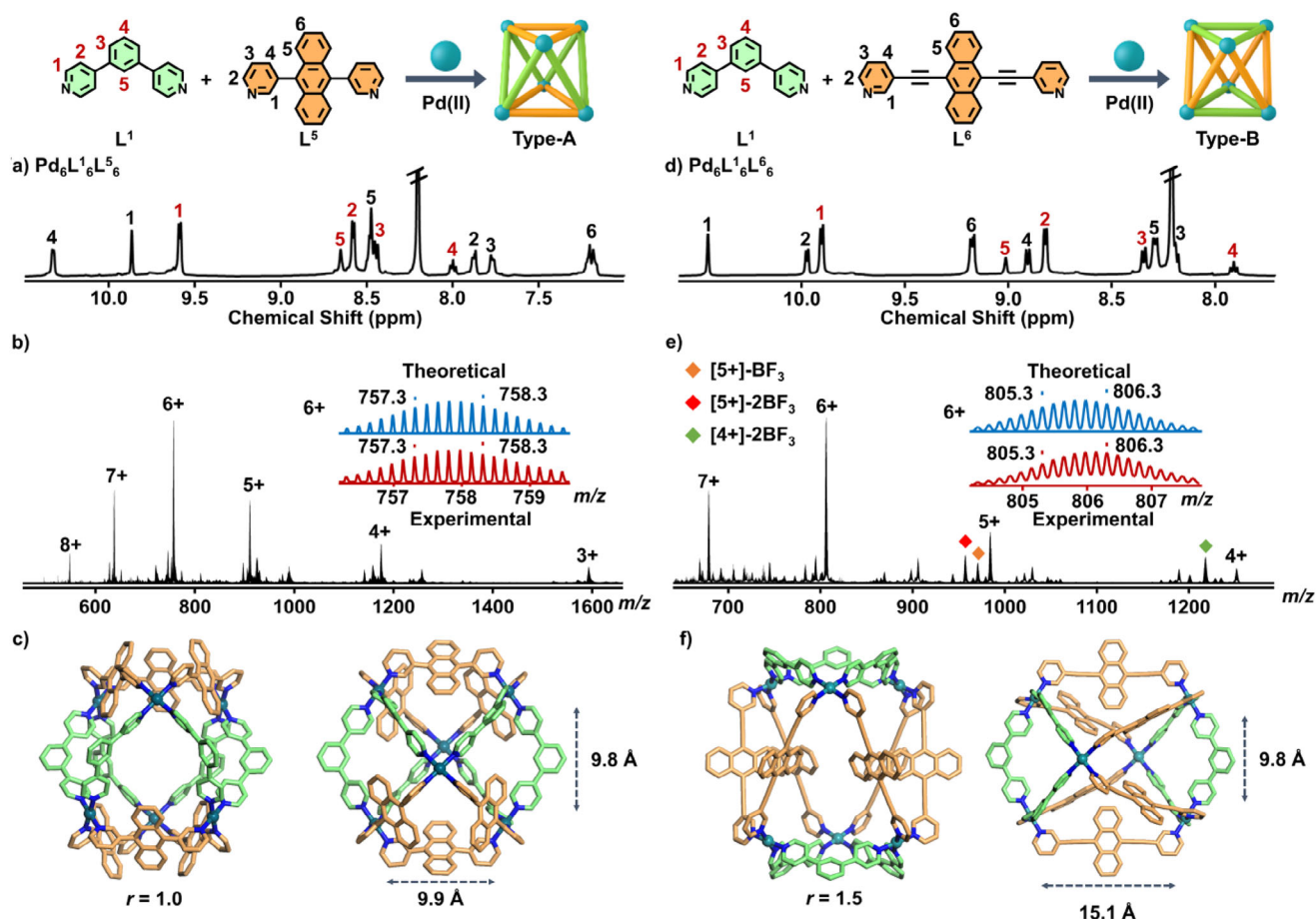


Figure 4. The assembled complex of L^1 , L^5 , and Pd(II) adopts the Type-A configuration; the assembled complex of L^1 , L^6 , and Pd(II) adopts the Type-B configuration. ^1H NMR spectra (500 MHz, 300 K) of a) $\text{Pd}_6\text{L}^1_6\text{L}^5_6$ and d) $\text{Pd}_6\text{L}^1_6\text{L}^6_6$. ESI-MS spectra of b) $\text{Pd}_6\text{L}^1_6\text{L}^5_6$ and e) $\text{Pd}_6\text{L}^1_6\text{L}^6_6$. The single-crystal structures of c) $\text{Pd}_6\text{L}^1_6\text{L}^5_6$ and f) $\text{Pd}_6\text{L}^1_6\text{L}^6_6$ (hydrogen atoms are omitted for clarity).^[44]

bridges linking the upper and lower triangular $[\text{Pd}_3(\text{L}^5)_3]^{6+}$ macrocyclic fragments. This preliminary data indicates that introducing bulky substituents into the ligand does not lead to configuration modulation of the heteroleptic cage. Notably, due to steric hindrance between hydrogen atoms on the anthryl ring and adjacent pyridyl rings, the anthryl ring in L^5 is perpendicular to the adjacent pyridyl ring rather than coplanar.

To further verify whether the introduction of bulky substituents would lead to configurational changes in the Type-B cage, a mixture of L^1 (1 equiv), L^6 (1 equiv), and $\text{Pd}(\text{CH}_3\text{CN})_4(\text{BF}_4)_2$ (1 equiv) was equilibrated in DMF-d_7 and reacted at 70 °C for 8 h, resulting in the clean formation of a heteroleptic cage $\text{Pd}_6\text{L}^1_6\text{L}^6_6$, as supported by ^1H NMR and ESI-MS data (Figure 4d,e). By slowly diffusing tetrahydrofuran vapor into the DMF solution of $\text{Pd}_6\text{L}^1_6\text{L}^6_6$, we successfully obtained yellow crystals suitable for X-ray diffraction. As shown in Figure 4f, the configuration of $\text{Pd}_6\text{L}^1_6\text{L}^6_6$ is determined to be Type-B. Additionally, the introduction of the alkynyl spacer effectively alleviates the steric hindrance between anthracene and pyridine, resulting in an overall coplanar structure for ligand L^6 . Combining the experiments of both $\text{Pd}_6\text{L}^1_6\text{L}^5_6$ and $\text{Pd}_6\text{L}^1_6\text{L}^6_6$, we have determined that steric elements do not alter the rule of

achieving cages configurational modulation based on ligand length ratio.

Conclusion

In summary, we proposed a strategy based on tuning the ligand length ratio to achieve configurational modulation of heteroleptic cages and successfully synthesized a series of heteroleptic $\text{Pd}_6\text{L}_6\text{L}'_6$ cages with two distinct configurations. Despite the theoretical existence of 48 possible configurations for the octahedral geometry of $\text{Pd}_6\text{L}_6\text{L}'_6$, our results demonstrate that the ligand length ratio can effectively direct the system toward a preferred configuration, enabling controllable modulation between Type-A and Type-B configurations. Specifically, when $r = 1.0$, the Type-A configuration is formed; whereas when r is adjusted to 1.4, 1.5, or 1.7, a Type-B configuration is formed. Furthermore, the single-crystal structures of $\text{Pd}_6\text{L}^1_6\text{L}^5_6$ and $\text{Pd}_6\text{L}^1_6\text{L}^6_6$ confirm that the introduction of bulky steric groups does not induce configuration changes, which further illustrates that the ligand length ratio is the primary factor governing configuration adjustment. This achievement marks a significant advance in the design principles of heteroleptic cage.

Supporting Information

Synthetic details and characterizations of ligands and metallo-complexes including NMR, ESI – MS and single-crystal data (PDF). The authors have cited additional references within the Supporting Information.

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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